Porush Yadav

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Education

- **MS, Mathematics and Scientific Computing** — IIT Kanpur CPI: 9.8 / 10 **2021**
- **BS, Mathematics and Scientific Computing** — IIT Kanpur CPI: 7.8 / 10 **2021**

Work Experience

Axis Bank Limited — IT:BUSINESS SYSTEM ANALYST **Aug 2021 – Nov 2021**

- Worked on an image-recognition system to automate document classification and data extraction.
- Evaluated its performance on diverse documents to identify and troubleshoot technical anomalies.

Samsung R&D Institute India — Intern **May 2020 – Jul 2020**

- Studied IQ data compression techniques (Vector Quantization, bit-vector compression, linear predictive coding) to minimize EVM values in uplink and downlink.
- Evaluated compression techniques using mathematical models and input data properties of IQ data.

eJET Systems — ML Intern **May 2019 – Jul 2019**

- Built an ML hotel chatbot: GRU intent classifier (Keras), NLTK extraction of food, quantity, attributes.
- Built a food recommender using K-Nearest Neighbors (KNN) search over word embeddings.

EZ Technologies — ML Intern **Dec 2018 – Jan 2019**

- Developed a real-time face-recognition attendance system using OpenCV and a pre-trained DNN to detect multiple student faces, with attendance records managed in MongoDB.

Projects

PDF Study Assistant (RAG) | Python, LangChain, FAISS, BM25, FastAPI, Gemini/Groq **Oct 2026**

- Architected an end-to-end RAG pipeline for a 193-page report, delivering accurately cited answers and strict out-of-domain refusals using local MiniLM embeddings and FAISS.
- Designed an evaluation harness (hit@k sweep) on gold-labeled queries, achieving 100% factual retrieval; boosted multi-page answer coverage from 42% to 73% by implementing hybrid (dense + BM25) retrieval.
- Shipped a resilient FastAPI chat interface featuring interactive PDF citations, automatic model fallback, rate-limit backoff, and output citation verification.

Consumer Complaint Analytics | Python, pandas, Dask, scikit-learn, Parquet, Streamlit **July 2026**

- Processed an 8.8 GB (15.2M rows) CFPB database within RAM limits using a Dask & Parquet pipeline.
- Trained a TF-IDF & Logistic Regression NLP classifier on 3.3M narratives: 87% accuracy (vs. 60.6% baseline).
- Uncovered a surge in templated filings (to 77%) via Cramér's V and chi-square tests; deployed on Streamlit.

Financial Market Surveillance | Python, pandas, NumPy, scikit-learn, yfinance, Jupyter **June 2026**

- Built an Isolation Forest anomaly-detection pipeline monitoring 5,800+ days of USD–INR exchange rates.
- Mapped 74.3% of flagged anomalies to known crises (2008 GFC, 2020 COVID) using rolling-volatility features.
- Developed a Market Stress Index capturing a 5.09× baseline volatility spike during peak historical stress.

European Option Pricing under Black-Scholes (MS Project) **Aug 2020 – Nov 2020**

- Analyzed risk-neutral valuation and delta-hedging of vanilla and binary European options via Black-Scholes framework, including option Greeks and implied volatility. Simulated volatility surface for MSFT stock in R.

Relevant Coursework

- **Computer Science:** Machine Learning, Computational Cognitive Science, Data Structures & Algorithms
- **Statistics:** Time Series Analysis, Stochastic Processes, Statistical Inference, Data Analysis, Sampling Theory
- **Maths:** Financial Mathematics, ANN/ML Approach for Differential Equations, Linear Algebra, Calculus

Technical Skills

- **Languages:** Python, C++, C, SQL, R, MATLAB
- **AI & Generative AI:** PyTorch, LangChain, RAG, FAISS, HuggingFace, LLM APIs (Gemini, Groq), BM25
- **Data Science & ML:** NLP, scikit-learn, pandas, NumPy, TensorFlow, Keras, NLTK, Dask, SHAP
- **Frameworks & Tools:** FastAPI, Streamlit, Parquet

Certifications

- Machine Learning Specialization; AI Code Review; Applying Machine Learning to your Data with GCP

Achievements & Scholarships

- **NTSE** Scholarship and **INSPIRE** Scholarship for Higher Education (SHE), Government of India